

NAME: _____

CLASS: 11MTA___ or 11MTX___

Circle your Teacher's name

Sing	Li	Saxena	Saw	Batra
Singh	Pathuis	Tolhurst	Iacona	Lees

CHERRYBROOK TECHNOLOGY HIGH SCHOOL



2019

YEAR 11

AP2

Mathematics Advanced

General

Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and calculations

Total Marks:

75

Section I – 10 marks (pages 2–5)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 65 marks (pages 6–22)

- Attempt Questions 11–27
- Allow about 1 hour and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1 Which of the following is equivalent to $\frac{3}{\sqrt{2}}$ with a rational denominator?

A $\frac{\sqrt{6}}{4}$

B $\frac{\sqrt{6}}{4}$

C $\frac{3\sqrt{2}}{2}$

D $\frac{3\sqrt{2}}{4}$

2 Which of the following is the closest to $\frac{2\pi}{7}$ in degrees?

A 0.9°

B 102.9°

C 51.4°

D 161.6°

3 It is given that $\ln a = \ln b + \ln c - \ln d$.

Which of the following statements is true?

A $a = b + c - d$

B $a = \frac{bc}{d}$

C $\ln a = \frac{\ln b \times \ln c}{\ln d}$

D $\ln a = \frac{b+c}{d}$

4 The solution to $|x - 3| = 6$ is

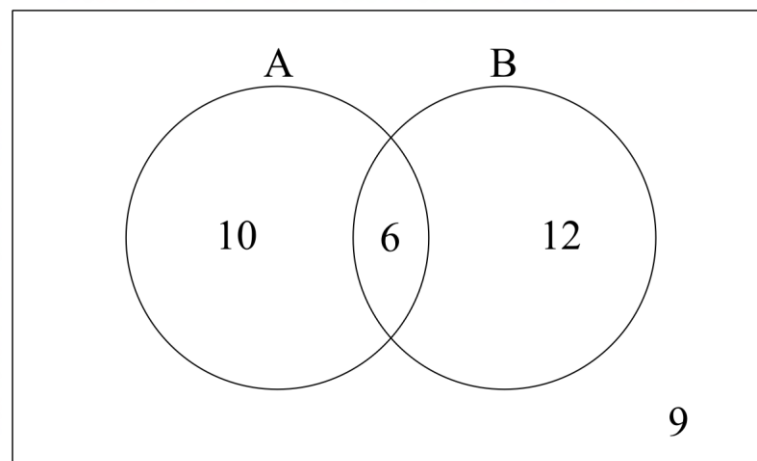
A $x = -3, 9$

B $x = 3, -9$

C $x = -3, -9$

D $x = 3, 9$

5 Consider the Venn diagram below.



According to the diagram, what is $P(A|B)$?

A $\frac{1}{3}$

B $\frac{5}{6}$

C $\frac{28}{37}$

D $\frac{3}{5}$

6 Which one of the following functions is *not* one-to-one?

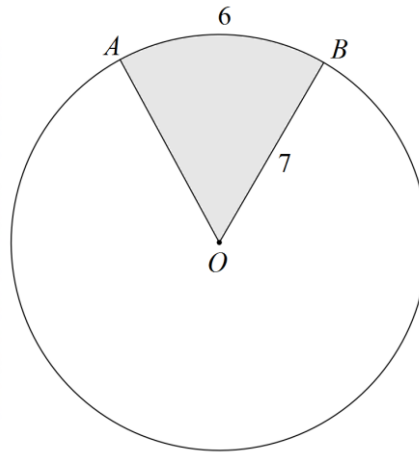
A $f(x) = 4 - x^2, x \geq 0$

B $f(x) = \sqrt{4 - x^2}$

C $f(x) = 1 - 4x$

D $f(x) = \frac{4}{x}$

- 7 The circle centred at O has a radius of 7. Arc AB has length 6, as shown in the diagram.



NOT TO
SCALE

What is the area of the shaded sector OAB ?

- A** 21π
- B** 21
- C** $\frac{343\pi}{12}$
- D** $\frac{343}{12}$
- 8 Given the equation $2x^2 - x + 5 = 0$ which statement is correct.
- A** The equation has equal roots.
- B** The equation has no real roots.
- C** The equation has unequal roots.
- D** The equation has rational roots.

- 9 The probability distribution of a random variable X is shown below:

x	0	1	2	3	4
$P(X = x)$	k	$2k$	$3k$	$2k$	k

What is the value of k ?

- A $\frac{1}{10}$
- B $\frac{1}{9}$
- C $\frac{1}{8}$
- D $\frac{1}{5}$
- 10 Which of the following is true if $f(x) = \sqrt{x}$ and $g(x) = 16 - x^2$
- A Domain of $f(g(x))$ is $[0,4]$ and range of $f(g(x))$ is $[0,4]$
- B Domain of $f(g(x))$ is $[-4,4]$ and range of $f(g(x))$ is $(0,4)$
- C Domain of $f(g(x))$ is $(-4,4)$ and range of $f(g(x))$ is $[0,4]$
- D Domain of $f(g(x))$ is $[-4,4]$ and range of $f(g(x))$ is $[0,4]$

Mathematics Advanced
Section II Answer Booklet

65 marks

Attempt Questions 11–27

Allow about 1 hour and 45 minutes for this section

Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
 - Your responses should include relevant mathematical reasoning and/or calculations.
 - Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
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Question 11 (5 marks)

If $f(x) = x^2 + 2x - 8$

- (a) Find the x and y intercepts. **2**

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- (b) Find the axis of symmetry. **1**

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- (c) Find the turning point. State whether it is a maximum or a minimum. **2**

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Question 12 (6 marks)

The volume V of a gas varies inversely with the pressure P it is under.

- (a) If the volume of the gas is 120 cm^3 when it is under a pressure of 40 kg/cm^3 , find an equation that relates the variables V and P . **2**

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- (b) What amount of pressure is needed so that the volume of the gas is 80 cm^3 ? **1**

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- (c) Draw a graph of the relationship between V and P , using P as the independent variable and V as the dependent variable, where $P > 0$. Label the axes. **2**



- (d) Describe what would happen to the volume of the gas if the pressure approaches zero. **1**

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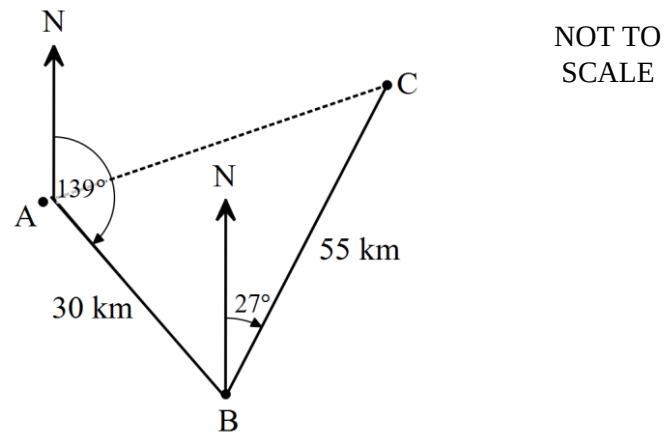
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Question 13 (4 marks)

A boat sails 30 kilometres from Island A to Island B on a bearing of 139° . It then sails 55 kilometres to Island C on a bearing of 27° from Island B.



Calculate the distance and bearing the boat must sail on to return to Island A from Island C. **4**

[illegible]

Question 14 (2 marks)

Given that $\cos \theta = -\frac{5}{8}$ and $0 < \theta < \pi$, find the value of $\tan \theta$.

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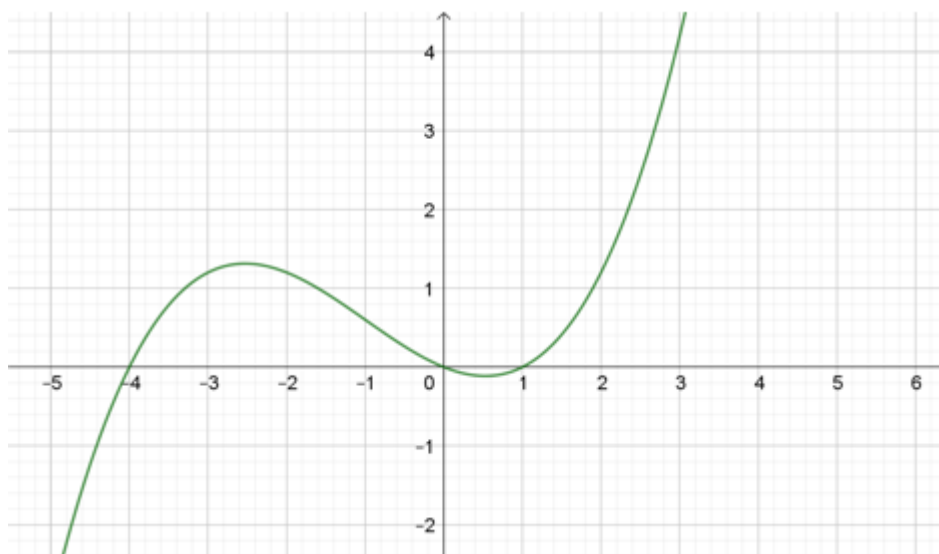
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Question 15 (2 marks)

The graph of $y = f(x)$ is given below. Sketch the graph of $y = -f(-x)$

2

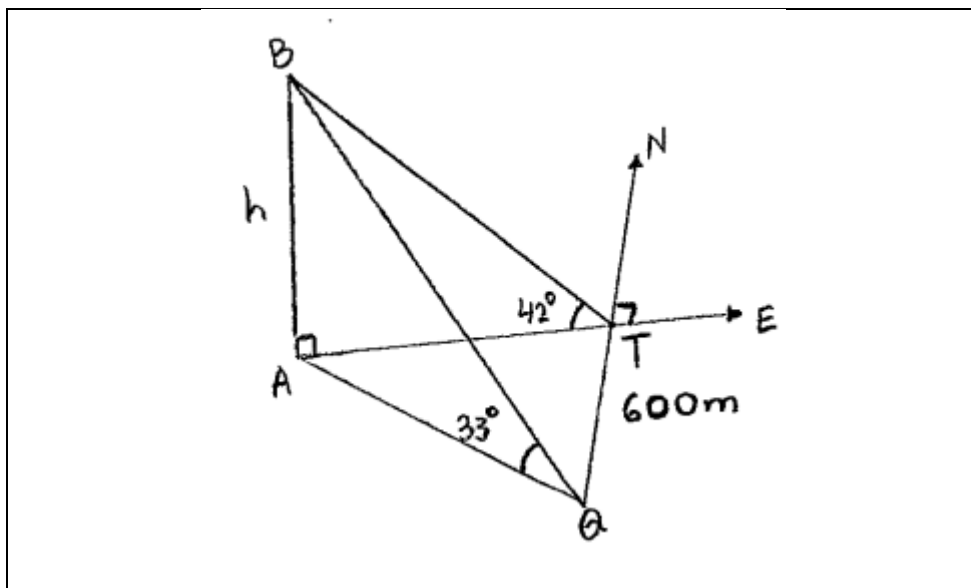


Question 16 (4 marks)

4

The elevation of a building from T due east of it is 42° ; at a place Q due south of T, the elevation of the building is 33° . The distance from T to Q is 600 metres.

Let the height (h) of the building be AB where (A) is the foot of the building. Determine the height of the building to 2 decimal places



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Question 17 (4 marks)

For the given discrete probability distribution

X	1	2	3	4
$P(X = x)$	0.3	0.2	0.4	0.1

- (a) Find the expected value. 2

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- (b) Find the variance. 2

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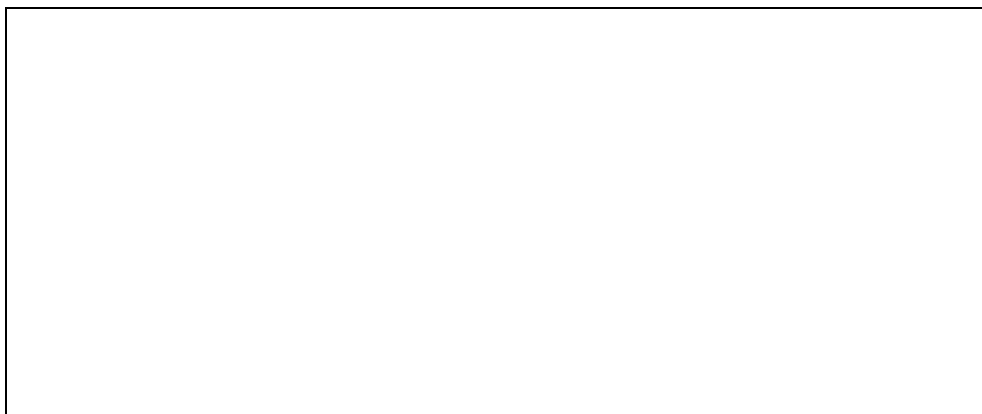
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Question 18 (3 marks)

In a class of 30 students, 13 students are in the soccer club, 14 students are in the chess club, and 8 students don't belong to either club.

- (a) Construct a Venn diagram to display this information.

2



- (b) If a student is chosen at random, what is the probability that the student belongs to both the clubs?

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Question 19 (2 marks)

John enters a biathlon which includes cross-country skiing and rifle shooting. He has a 39% chance of winning the cross-country skiing and a 32% chance of winning both races. Find the probability that he wins the rifle shooting, if he has won the cross-country skiing race?

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(The **biathlon** is a winter sport that combines cross-country skiing and rifle shooting. It is treated as a race where the contestant with the shortest total time wins.)

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Question 20 (6 marks)

- (a) Using the method of first principles, find $f'(x)$ if $f(x) = 3x^2 - 5$ **3**

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- (b) Hence, find the equation of the line that is tangent to $f(x)$ at $x = 2$. **3**

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Question 21 (6 marks)

Differentiate the following expressions with respect to x .

(a) $2x^2 + \frac{5}{x}$ 2

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(b) $(3x^2 - 5x)^5$ 2

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(c) $(3x - 4)(4x + 5)$ 2

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Question 22 (4 marks)

If $y = \frac{2x+4}{\sqrt{x+1}}$, Show that $\frac{dy}{dx} = \frac{x}{\sqrt{(1+x)^3}}$

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Question 23 (3 marks)

Velocity, V metres per second, of a particle at time t seconds is given by **3**

$$V = \frac{8t^3}{3} - 8t^2 + 12t + 8$$

Show that the acceleration is never less than 4 metres per second per second.

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Question 24 (2 marks)

Differentiate $y = -2e^{-3x} + 5x^3$ **2**

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Question 25 (3 marks)

The amount A of money in a bank account after n years grows with compound interest according to the formula $A = 820(1.035)^n$. **3**

Find how many years it will take for the amount in the bank to be \$2000?

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Question 26 (4 marks)

The pH value of a solution is given by the formula $\text{pH} = -\log_{10}[\text{H}^+]$, where $[\text{H}^+]$ is the concentration of hydrogen ions in moles per litre.

- (a) Find the pH value of a solution in a chemistry lab whose label indicates its concentration of hydrogen ions is 2.5×10^{-4} moles per litre. **1**
Give your answer to one decimal place.

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- (b) Find the concentration of hydrogen ions in moles per litre for vinegar if a test strip indicates it has a pH of 2.4. **3**
Give your answer as $a \times 10^b$ to two significant figures.

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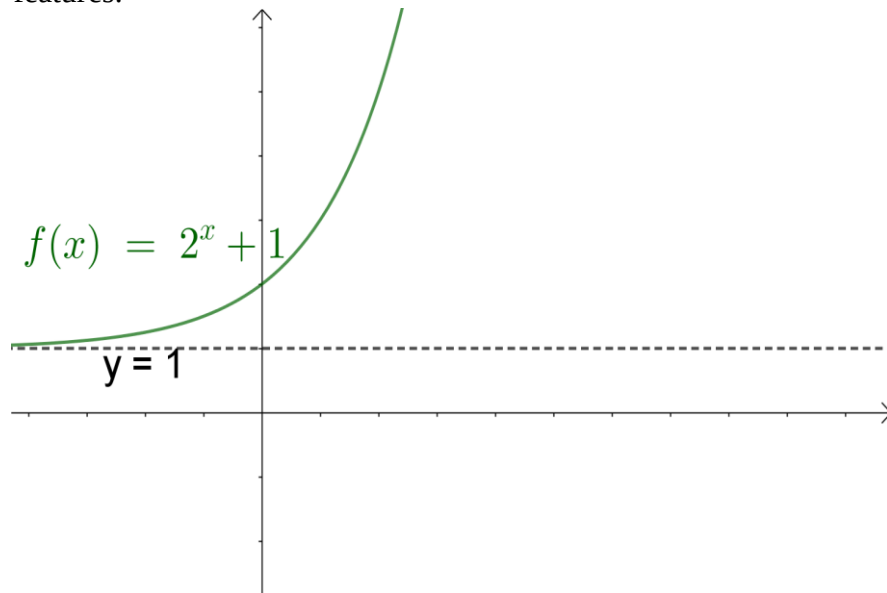
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Question 27 (5 marks)

(a) The graph of $f(x) = 2^x + 1$ is given below.

3

On the same set of axis sketch the graph of $g(x) = \log_2(x - 1)$ showing key features.



(b) Hence, state the domain and range of $g(x)$.

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End of examination

[illegible]

Section II extra writing space

If you use this space, clearly indicate which question you are answering

SOLUTIONS

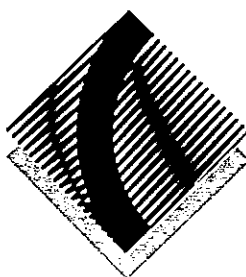
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Section I

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- 1 Which of the following is equivalent to $\frac{3}{\sqrt{2}}$ with a rational denominator?

A $\frac{\sqrt{6}}{4}$

B $\frac{\sqrt{6}}{4}$

C $\frac{3\sqrt{2}}{2}$

D $\frac{3\sqrt{2}}{4}$

$$\frac{3}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{3\sqrt{2}}{2}$$

- 2 Which of the following is the closest to $\frac{2\pi}{7}$ in degrees?

A 0.9°

B 102.9°

C 51.4°

D 161.6°

$$\frac{2\pi}{7} = 51.4^\circ$$

- 3 It is given that $\ln a = \ln b + \ln c - \ln d$.

Which of the following statements is true?

A $a = b + c - d$

B $a = \frac{bc}{d}$

C $\ln a = \frac{\ln b \times \ln c}{\ln d}$

D $\ln a = \frac{b+c}{d}$

$$\ln a = \ln \frac{bc}{d}$$

$$a = \frac{bc}{d}$$

4 The solution to $|x-3|=6$ is

A $x = -3, 9$

B $x = 3, -9$

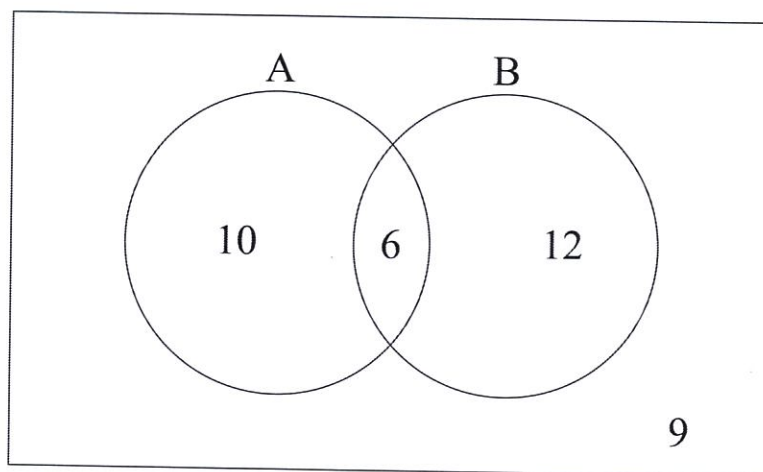
C $x = -3, -9$

D $x = 3, 9$

$$\begin{aligned} x-3 &= 6 \\ x &= 9 \end{aligned}$$

$$\begin{aligned} \text{or } -(x-3) &= 6 \\ x-3 &= -6 \\ x &= -6+3 \\ &= -3 \end{aligned}$$

5 Consider the Venn diagram below.



According to the diagram, what is $P(A|B)$?

A $\frac{1}{3}$

B $\frac{5}{6}$

C $\frac{28}{37}$

D $\frac{3}{5}$

$$\frac{P(A \cap B)}{P(B)} = \frac{6}{18} = \frac{1}{3}$$

6 Which one of the following functions is *not* one-to-one?

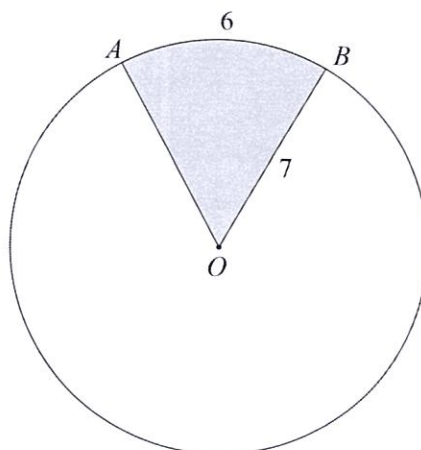
A $f(x) = 4 - x^2, x \geq 0$

B $f(x) = \sqrt{4 - x^2}$

C $f(x) = 1 - 4x$

D $f(x) = \frac{4}{x}$

- 7 The circle centred at O has a radius of 7. Arc AB has length 6, as shown in the diagram.



NOT TO
SCALE

What is the area of the shaded sector OAB ?

A 21π

B 21

C $\frac{343\pi}{12}$

D $\frac{343}{12}$

$$\begin{aligned} l &= r\theta \\ 6 &= \theta \times 7 \\ \theta &= \frac{6}{7} \\ A &= \frac{1}{2} r^2 \times \theta \\ &= \frac{1}{2} \times 7^2 \times \frac{6}{7} \\ &= 21 \end{aligned}$$

- 8 Given the equation $2x^2 - x + 5 = 0$ which statement is correct.

A The equation has equal roots.

B The equation has no real roots.

C The equation has unequal roots.

D The equation has rational roots.

$$\begin{aligned} \Delta &= b^2 - 4ac \\ &= (-1)^2 - 4 \times 2 \times 5 \\ &= 1 - 40 \\ &= -39 \\ \therefore &\text{ No real roots} \end{aligned}$$

- 9 The probability distribution of a random variable X is shown below:

x	0	1	2	3	4
$P(X=x)$	k	$2k$	$3k$	$2k$	k

What is the value of k ?

A $\frac{1}{10}$

B $\frac{1}{9}$

C $\frac{1}{8}$

D $\frac{1}{5}$

$$k + 2k + 3k + 2k + k = 1$$

$$9k = 1$$

$$k = \frac{1}{9}$$

- 10 Which of the following is true if $f(x) = \sqrt{x}$ and $g(x) = 16 - x^2$

A Domain of $f(g(x))$ is $[0,4]$ and range of $f(g(x))$ is $[0,4]$

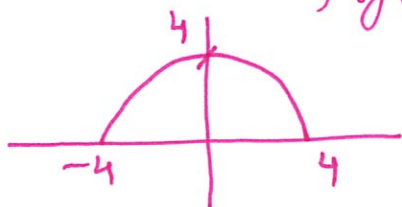
B Domain of $f(g(x))$ is $[-4,4]$ and range of $f(g(x))$ is $(0,4)$

C Domain of $f(g(x))$ is $(-4,4)$ and range of $f(g(x))$ is $[0,4]$

D Domain of $f(g(x))$ is $[-4,4]$ and range of $f(g(x))$ is $[0,4]$

$$f(g(x)) = f(16 - x^2)$$

$$f(g(x)) = \sqrt{16 - x^2}$$



Domain: $[-4,4]$
Range: $[0,4]$

Mathematics Advanced
Section II Answer Booklet

70 marks

Attempt Questions 11–27

Allow about 1 hour and 45 minutes for this section

Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
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-

Question 11 (5 marks)

If $f(x) = x^2 + 2x - 8$

- (a) Find the x and y intercepts.

2

y-intercept when $x = 0$
 $f(0) = 0^2 + 2 \times 0 - 8$
 $= -8$

x-intercept when $y = 0$

$$x^2 + 2x - 8 = 0$$
$$(x+4)(x-2) = 0$$
$$x = 2 \text{ or } x = -4$$

- (b) Find the axis of symmetry.

1

$$x = -\frac{b}{2a}$$

$$x = \frac{-2}{2 \times 1}$$

$$x = -1$$

- (c) Find the turning point. State whether it is a maximum or a minimum.

2

Turning point on the axis of symmetry

$$f(-1) = (-1)^2 + 2(-1) - 8$$
$$= 1 - 2 - 8$$
$$= -9$$

Since $a > 0 \therefore$ parabola is concave up
Turning pt. is a minimum at $(-1, -9)$

Question 12 (6 marks)

The volume V of a gas varies inversely with the pressure P it is under.

- (a) If the volume of the gas is 120 cm^3 when it is under a pressure of 40 kg/cm^3 , find an equation that relates the variables V and P . 2

$$V = \frac{K}{P} \quad \therefore K = 4800$$

$$120 = \frac{K}{40} \quad V = \frac{4800}{P} \text{ or } V \times P = 4800$$

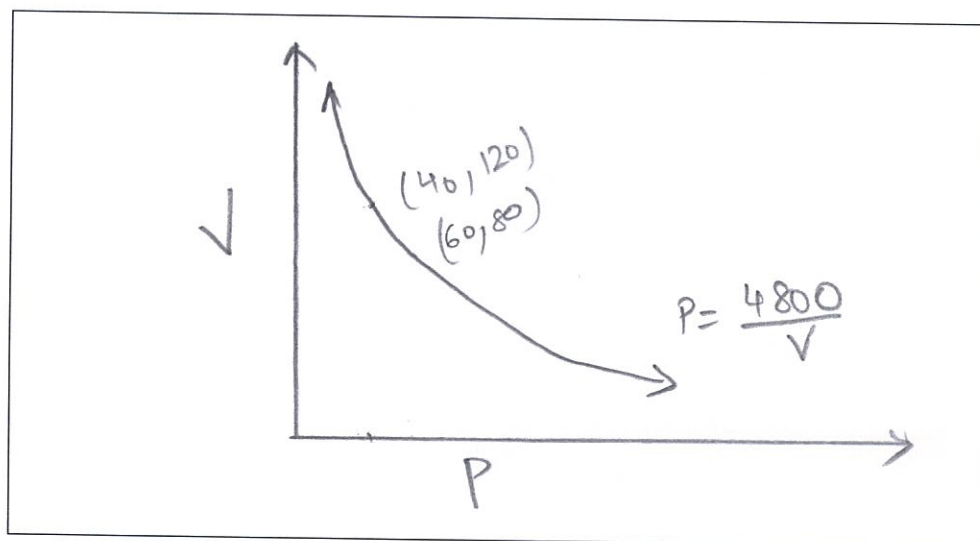
$$\text{or } P = \frac{4800}{V}$$

- (b) What amount of pressure is needed so that the volume of the gas is 80 cm^3 ? 1

$$P = \frac{4800}{80}$$

$$= 60 \text{ kg/cm}^3$$

- (c) Draw a graph of the relationship between V and P , using P as the independent variable and V as the dependent variable, where $P > 0$. Label the axes. 2

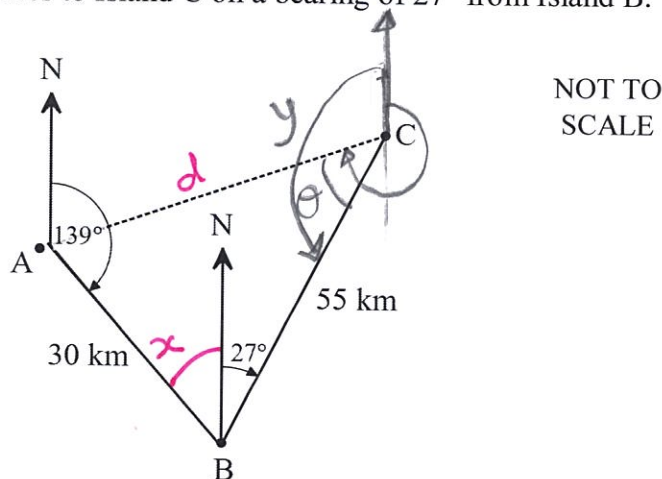


- (d) Describe what would happen to the volume of the gas if the pressure approaches zero. 1

As $P \rightarrow 0$, $V \rightarrow \infty$

Question 13 (4 marks)

A boat sails 30 kilometres from Island A to Island B on a bearing of 139° . It then sails 55 kilometres to Island C on a bearing of 27° from Island B.



Calculate the distance and bearing the boat must sail on to return to Island A from Island C. 4

$$x = 41^\circ \text{ (Co-int } \angle\text{s on parallel lines are supplementary)}$$

$$\therefore \angle ABC = 41 + 27 = 68^\circ$$

$$d = \sqrt{30^2 + 55^2 - 2 \times 30 \times 55 \times \cos 68^\circ}$$

$$d = 51.8536 \approx 51.85$$

$$\frac{\sin \theta}{30} = \frac{\sin 68}{55}$$

$$\sin \theta = \frac{30 \times \sin 68}{55}$$

$$\theta \approx 32^\circ$$

$$y = 180 - 27^\circ \text{ (Co-int } \angle\text{s on } \parallel \text{ lines are supplementary)}$$

$$= 153^\circ$$

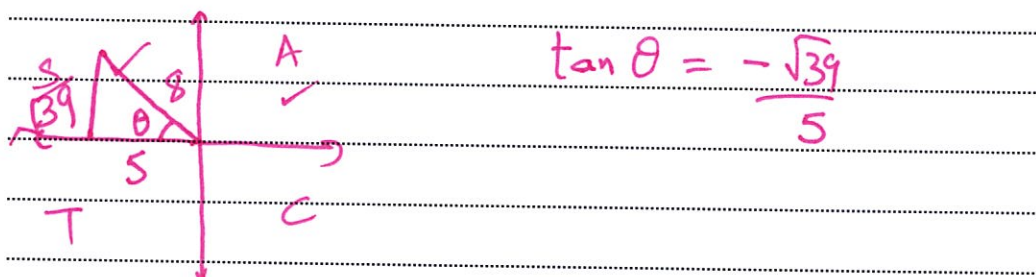
$$\text{The bearing of A from C is } 360^\circ - 153 + 32$$

$$= 239^\circ \text{ T}$$

Question 14 (2 marks)

Given that $\cos \theta = -\frac{5}{8}$ and $0 < \theta < \pi$, find the value of $\tan \theta$.

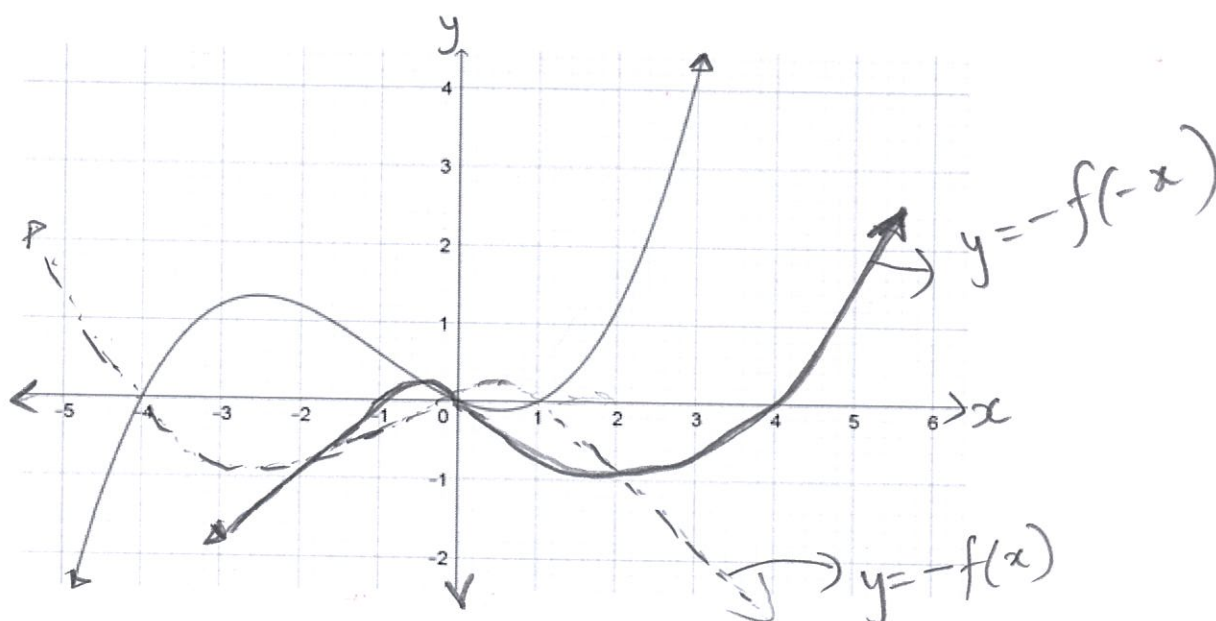
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Question 15 (2 marks)

The graph of $y = f(x)$ is given below. Sketch the graph of $y = -f(-x)$

2

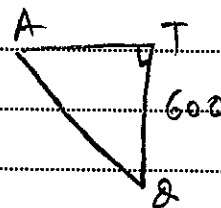
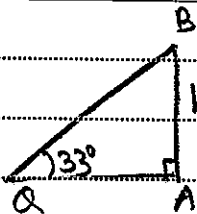
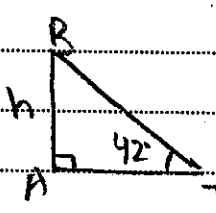
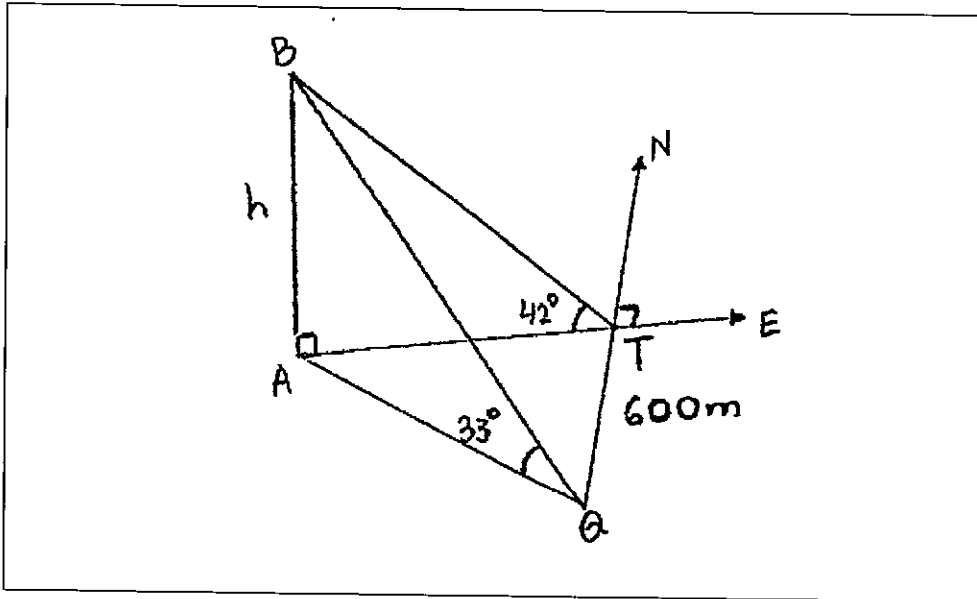


Question 16 (4 marks)

4

The elevation of a building from T due east of it is 42° ; at a place Q due south of T, the elevation of the building is 33° . The distance from T to Q is 600 metres.

Let the height (h) of the building be AB where (A) is the foot of the building. Determine the height of the building to 2 decimal places



$$\tan 42^\circ = \frac{h}{AT}$$

$$\tan 33^\circ = \frac{h}{AQ}$$

$$AT = \frac{h}{\tan 42^\circ}$$

$$AQ = \frac{h}{\tan 33^\circ}$$

$$AQ^2 - AT^2 = 600^2$$

$$\frac{h^2}{\tan^2 33^\circ} - \frac{h^2}{\tan^2 42^\circ} = 600^2$$

$$h^2 \left[\frac{1}{\tan^2 33^\circ} - \frac{1}{\tan^2 42^\circ} \right] = 600^2$$

$$h^2 \times \left[\frac{\tan^2 42^\circ - \tan^2 33^\circ}{\tan^2 33^\circ \times \tan^2 42^\circ} \right] = 600^2$$

$$h^2 = \frac{600^2 \times \tan^2 33^\circ \times \tan^2 42^\circ}{\tan^2 42^\circ - \tan^2 33^\circ}$$

$$h = \sqrt{\frac{600^2 \times \tan^2 33^\circ \times \tan^2 42^\circ}{\tan^2 42^\circ - \tan^2 33^\circ}} = 562.51 \text{ m}$$

Question 17 (4 marks)

For the given discrete probability distribution

X	1	2	3	4
$P(X = x)$	0.3	0.2	0.4	0.1

(a) Find the expected value.

2

$$E(X) = \sum x p(x)$$

$$= 1 \times 0.3 + 2 \times 0.2 + 3 \times 0.4 + 4 \times 0.1$$

$$= 0.3 + 0.4 + 1.2 + 0.4$$

$$= 2.3$$

(b) Find the variance.

2

$$\text{Var}(X) = 1^2 \times 0.3 + 2^2 \times 0.2 + 3^2 \times 0.4 + 4^2 \times 0.1 - (2.3)^2$$

$$= 1.01$$

or

X	1	2	3	4
$(x - \mu)$	-1.3	-0.3	0.7	1.7
$P(X = x)$	0.3	0.2	0.4	0.1

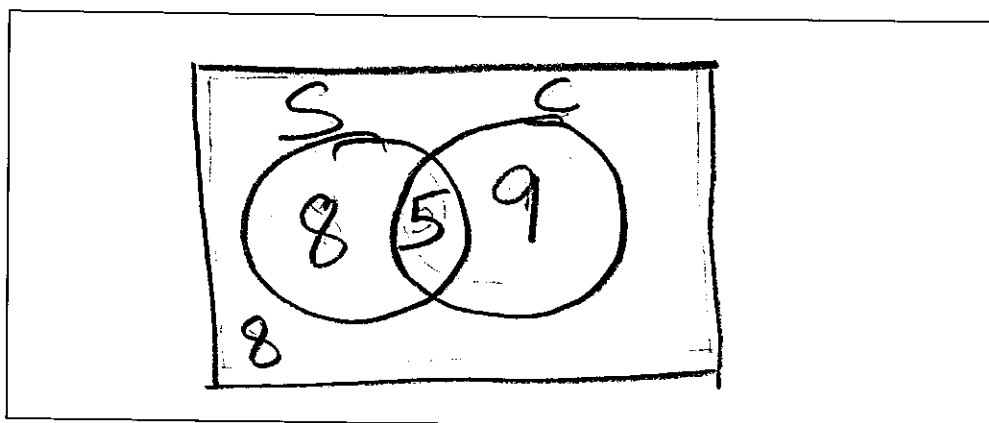
$$\text{Var}(X) = (-1.3)^2 \times 0.3 + (-0.3)^2 \times 0.2 + (0.7)^2 \times 0.4 + 1.7^2 \times 0.1$$
$$= 1.01$$

Question 18 (3 marks)

In a class of 30 students, 13 students are in the soccer club, 14 students are in the chess club, and 8 students don't belong to either club.

- (a) Construct a Venn diagram to display this information.

2



- (b) If a student is chosen at random, what is the probability that the student belongs to both the clubs?

1

$$\frac{5}{30} = \frac{1}{6}$$

Question 19 (2 marks)

John enters a biathlon which includes cross-country skiing and rifle shooting. He has a 39% chance of winning the cross-country skiing and a 32% chance of winning both races. Find the probability that he wins the rifle shooting, if he has won the cross-country skiing race?

2

(The **biathlon** is a winter sport that combines cross-country skiing and rifle shooting. It is treated as a race where the contestant with the shortest total time wins.)

$$P(\text{win x-ski}) = 0.39$$

$$P(\text{win both}) = 0.32$$

$$P(\text{win rifle} / \text{x-ski}) = \frac{P(\text{win both})}{P(\text{win x-ski})} = \frac{0.32}{0.39} = 0.82 \text{ or } 82\%$$

Question 20 (6 marks)

- (a) Using the method of first principles, find $f'(x)$ if $f(x) = 3x^2 - 5$

3

$$\begin{aligned}f(x+h) &= 3(x+h)^2 - 5 \\&= 3(x^2 + 2xh + h^2) - 5 \\&= 3x^2 + 6xh + 3h^2 - 5\end{aligned}$$

$$\begin{aligned}f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\&= \lim_{h \rightarrow 0} \frac{(3x^2 + 6xh + 3h^2 - 5) - (3x^2 - 5)}{h} \\&= \lim_{h \rightarrow 0} \frac{6xh + 3h^2}{h} = \lim_{h \rightarrow 0} \cancel{h}(6x + 3h) \\&= 6x\end{aligned}$$

- (b) Hence, find the equation of the line that is tangent to $f(x)$ at $x = 2$.

3

when $x = 2$

$$\begin{aligned}f(x) &= 3(2)^2 - 5 \\&= 7\end{aligned}$$

$$f'(x) = 6x$$

$$f'(2) = 12$$

$$y - 7 = 12(x - 2)$$

$$y = 12x - 17$$

Question 21 (6 marks)

Differentiate the following expressions with respect to x .

(a) $2x^2 + \frac{5}{x}$

2

$$\frac{d}{dx} \left(2x^2 + \frac{5}{x} \right)$$

$$= 4x - \frac{5}{x^2}$$

$$= 4x - \frac{5}{x^2}$$

(b) $(3x^2 - 5x)^5$

2

$$\frac{d}{dx} (3x^2 - 5x)^5$$

$$5(3x^2 - 5x)^4 \times (6x - 5)$$

(c) $(3x - 4)(4x + 5)$

2

$$\frac{d}{dx} (3x - 4)(4x + 5)$$

$$4(3x - 4) + (4x + 5) \times 3$$

$$12x - 16 + 12x + 15$$

$$24x - 1$$

Question 22 (4 marks)

If $y = \frac{2x+4}{\sqrt{x+1}}$. Show that $\frac{dy}{dx} = \frac{x}{\sqrt{(1+x)^3}}$

4

$$u = 2x+4$$

$$v = \sqrt{x+1}$$

$$u' = 2$$

$$v' = \frac{1}{2\sqrt{x+1}}$$

$$y' = \frac{vu' - uv'}{v^2}$$

$$= \frac{2\sqrt{x+1} - \frac{(2x+4)}{2\sqrt{x+1}}}{(x+1)}$$

$$= \frac{2\sqrt{x+1} - \frac{2(x+2)}{2\sqrt{x+1}}}{(x+1)}$$

$$= \frac{2(\sqrt{x+1})^2 - (x+2)}{(x+1)\sqrt{x+1}} = \frac{2(x+1) - (x+2)}{\sqrt{(x+1)^3}}$$

$$= \frac{x}{\sqrt{(x+1)^3}}$$

Question 23 (3 marks)

Velocity, V metres per second, of a particle at time t seconds is given by

3

$$V = \frac{8t^3}{3} - 8t^2 + 12t + 8$$

Show that the acceleration is never less than 4 metres per second per second.

$$V = \frac{8t^3}{3} - 8t^2 + 12t + 8$$

$$a = \frac{dV}{dt} = \frac{8}{3} \times 3t^2 - 16t + 12$$

$$a = 8t^2 - 16t + 12$$

$$\text{Axis of symmetry } t = \frac{-b}{2a} = \frac{-(-16)}{2 \times 8}$$

$$t = 1$$

Vertex. $\rightarrow (1, 4)$ Since minimum turning point value of

$$a = 8(1)^2 - 16 \times 1 + 12$$
$$\underline{a = 4.}$$

Question 24 (2 marks)

Differentiate $y = -2e^{-3x} + 5x^3$

2

$$\frac{dy}{dx} = -2e^{-3x} \times (-3) + 15x^2$$

$$= 6e^{-3x} + 15x^2$$

Question 25 (3 marks)

The amount A of money in a bank account after n years grows with compound interest according to the formula $A = 820(1.035)^n$.
Find how many years it will take for the amount in the bank to be \$2000?

$$A = 820(1.035)^n$$

$$2000 = 820(1.035)^n$$

$$1.035^n = \frac{2000}{820}$$

$$n = \frac{\log 2000}{\log 1.035 \cdot 820}$$

$$= \frac{\log_{10} 2000}{\log_{10} 820}$$

$$\log_{10} 1.035$$

$$= 25.9 \approx 26 \text{ year s.}$$

Question 26 (4 marks)

The pH value of a solution is given by the formula $\text{pH} = -\log_{10}[\text{H}^+]$, where $[\text{H}^+]$ is the concentration of hydrogen ions in moles per litre.

- (a) Find the pH value of a solution in a chemistry lab whose label indicates its concentration of hydrogen ions is 2.5×10^{-4} moles per litre. Give your answer to one decimal place. 1

$$\text{pH} = -\log_{10} 2.5 \times 10^{-4}$$

$$= 3.6$$

- (b) Find the concentration of hydrogen ions in moles per litre for vinegar if a test strip indicates it has a pH of 2.4. Give your answer as $a \times 10^b$ to two significant figures. 3

$$2.4 = -\log_{10}[\text{H}^+]$$

$$-2.4 = \log_{10}[\text{H}^+]$$

$$[\text{H}^+] = 10^{-2.4}$$

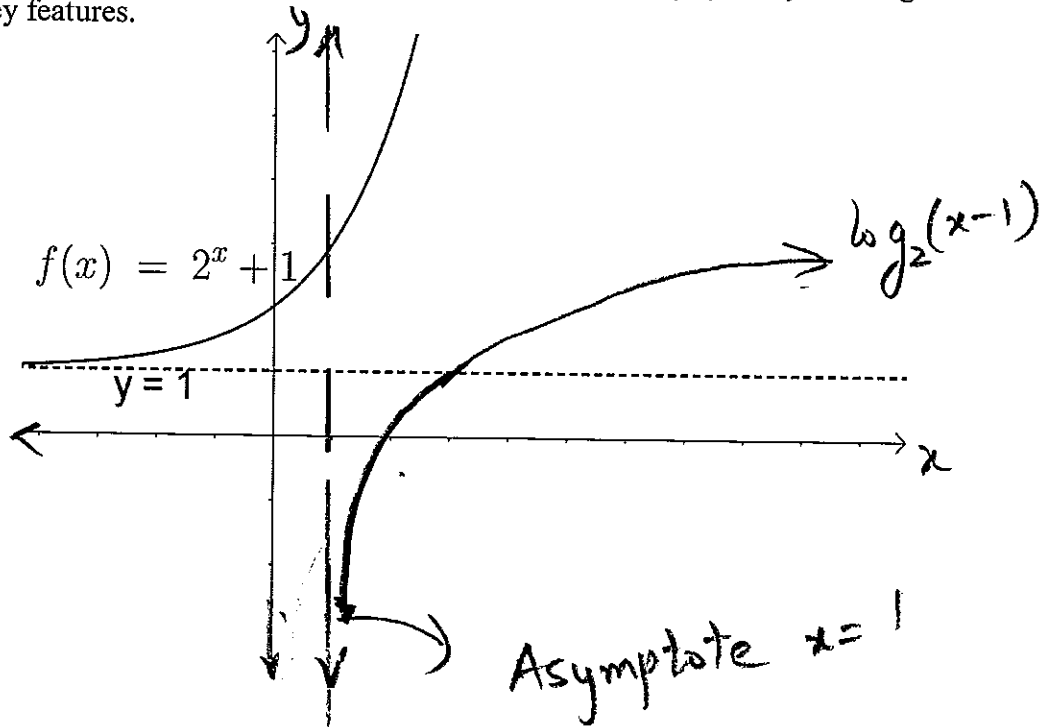
$$= 4.0 \times 10^{-3}$$

Question 27 (5 marks)

- (a) The graph of $f(x) = 2^x + 1$ is given below.

3

On the same set of axis sketch the graph of $g(x) = \log_2(x - 1)$ showing key features.



- (b) Hence, state the domain and range of $g(x)$.

2

Domain $\rightarrow (1, \infty)$
Range $\rightarrow (-\infty, \infty)$

End of examination